SUNY Old Westbury

Dept. OF Mathematics, Computers & Information Science

SCHOOL OF ARTS AND SCIENCE

COMPUTER SCIENCE PROGRAM

COURSE TITLE: Software Engineering

COURSE NUMBER: CS 4501 **MEETS:** Mon & Wed, 6:20-8:00pm **Room:** NAB 0101

CREDIT HOURS: 4

INSTRUCTOR: Doyoung Park, Ph.D

OFFICE: NAB 2008

PHONE: 516-628-5642

EMAIL: parkd@oldwestbury.edu

OFFICE HOURS: 4:20-5:50pm on Mon & Wed (or by an appointment)

TERM(S) OFFERED: Fall, 2024

TEXTBOOK:

(Required)

Title: Object-Oriented Software Engineering: Using UML,Patterns and Java
Author: Bernd Bruegge, Allen H. Dutoit
Publisher: Prentice Hall, Upper Saddle River, NJ, 2009
ISBN: ISBN-10:0136061257 OR ISBN-13: 978-0136061250

(Recommended)

1) Applying UML and Patterns - An Introduction to Object-Orient, Craig Larman, 2001, Prentice Hall, 2004, ISBN: 0-13-148906-2.
2) Principles of Object-Oriented Software Development, Anton Eliens, 1995, Addison-Wesley
3) Developing Software with UML, Object-Oriented Analysis and D, Bernd Oestereich, 1999, Addison-Wesley, QA76.9.03503713 1999
4) Object-Oriented Analysis and Design with Applications, 2nd e, G. Booch, 1994, Benjamin/Cummings, Redwood City, CA, QA76.64.B66 1994.
5) Object-Oriented Software Engineering, Timothy C. Lethbridge and Robert Laganiere, 2001, McGraw-Hill, ISBN: 0-07-709761-0

PREREQUISITES: CS 2511 (Computer Programming II), CS 3810 or CS 3611

COURSE DESCRIPTION:

Requirements definition, analysis and modeling including use cases and use case paths, domain models, state transition diagrams; techniques to increase robustness and avoid disastrous defects; object oriented architecture and design patterns and specifications in UML; performance impact of design choices; analysis of designs regarding maintainability and testability; security engineering; practical system test and glass box testing fundamentals; verification of test coverage via decision tables and state transition table.

COURSE OBJECTIVES (LERNING OUTCOMES):

Students who pass this class have the working knowledge of software engineering principles and hands-on experience in working on real life projects. At the end of the course student will possess the skills necessary to:
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- Understand the principles of software engineering outlined above.
- Understand and be fluent in the use of software engineering terminology and nomenclature
- Be capable of intelligently communicating with most members in a software development organization (management, analysts, architects, developers, ...)
- Be able to create and use planning, requirements analysis, domain analysis and design artifacts.
- Be capable of taking on the role of systems analyst in a software development organization
- Be able to document all phases of the software development processes
- Have the necessary tools to become a software development project manager
- They can design and implement complex software solutions using state of the art software engineering techniques.
- They have working knowledge of CASE tools, source control, and project management.
- They have deep knowledge of the technologies they used for implementing their project.
- They know how to test and document software.
- They are capable of working as part of a software team and develop significant projects

TOPICS:

1. To impart a firm understanding of what it means to be a software engineer
2. To discuss the differences between programs and engineered software produced via a software production process

3. To impart an understanding of software development lifecycles, the phases and activities of a lifecycle, and the artifacts (documents and code) created in each phase of a lifecycle
4. To present an overview of software management topics, including project planning tools and diagrams, team organization, human factors, configuration management, and risk analysis
5. To cover Requirements Analysis, including use cases, traditional requirements gathering techniques, and user interface prototypes as a tool for obtaining customer feedback
6. Explore UML class diagrams, state charts, and interaction diagrams for domain analysis, systems design, use-case analysis, modeling, etc.
7. Investigate software quality and software quality assurance (reviews, inspections, unit testing, integration testing, stress testing, system testing, etc.)
8. Explore issues of maintenance, reuse, and extendibility
9. Experience the application of specifications, software modeling an development tool, in an individual and collaborative team environment, toward software development

SOFTWARE:

These tools and frameworks will be mentioned at certain point during class and we will have a chance to deal with them based on the lecture topic.

- Lucid Chart https://www.lucidchart.com/pages/uml-use-case-diagram
- Open Source UML tools http://java-source.net/open-source/uml-modeling
- Tools for Requirements Managements
  (DOORS) http://www-01.ibm.com/software/rational/support/telelogic/
  (RequisitePro) https://www.ibm.com/products
  (RD-Link) http://www.ring-zero.com/
  (Unicase) http://www.eclipse.org/ecp/
- The Dresden OCL Toolkit http://dresden-ocl.sourceforge.net/
- Frameworks for Continuous Integration
  (CruiseControl) http://cruisecontrol.sourceforge.net/
  (Hudson from Kohsuke Kawaguchi)
  http://www.oracle.com/splash/java.net/maintenance/index.html

INTENDED AUDIENCE:

This is intended for students who major in Computer Science or Management Information Systems or related field.

INSTRUCTIONAL MODALITY(IES): Lecture/Discussion/Labs
ACTIVITIES/ASSIGNMENTS/REQUIREMENTS:

[Exam] There will be a midterm and a final exam. The final exam will focus on material covered after the midterm. Make-up exams will not be given without approval from the professor. Any unapproved absence from an exam will yield a score of zero, no exceptions.

[Assignments] Several homework assignments will be given. I will post the assignments on Blackboard and discuss them in class. For each assignment, If you submit *m* days later, your maximum grade will 100*(1-0.*m*). For example, if *m*=2, your maximum grade is 100*(10.2)=100*0.8=80.

[Quizzes] There will be certain number of pop quizzes during the semester. Those quizzes will be based on contents covered in previous lectures (lectures from the last quiz through the most recent lecture.)

ATTENDANCE:

Class attendance is required and a record of attendance will be kept. If you miss a class it is your responsibility to find out what material was covered in class, what the homework was and if any announcements have been made about the schedule for upcoming exams.

EVALUATION:

Letter grades will be assigned to each student based on a mathematical calculation of the points earned on the homework, projects, quizzes as well as middle and final examinations. The weights of the exams are:

Your final grade in the course will be based on the following:

Assignments (or Quizzes)	30%
Midterm exam	25%
Final exam	35%
Class Participation (and Attendance)	10%

<table>
<tr><td rowspan="3">A = [94, 100]
A⁻ = [90, 93]</td><td>B⁺ = [87, 89]</td><td>C⁺ = [77, 79]</td><td>D⁺ = [67, 69]</td><td rowspan="3">F = [0, 59]</td></tr>
<tr><td>B = [84, 86]</td><td>C = [74, 76]</td><td>D = [64, 66]</td></tr>
<tr><td>B⁻ = [80, 83]</td><td>C⁻ = [70, 73]</td><td>D⁻ = [60, 63]</td></tr>
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Note:

1) Any type of cheating/plagiarism in tests or projects will result in a grade of "F".
2) The teaching management tool ***Blackboard*** will be used to post the assignments, collect the submission, communicate and post grades.

TENTATIVE COURSE SCHEDULE:

Sessions / week	Topic	Reading
August Week 1	Getting Started: Introduction to Software Engineering	Chap 1
September Week 2	Modeling with UML (Part 1, Part 2)	Chap 2
Week 3	Modeling with UML (Review, UML 2 Highlights)	Chap 2
Week 4	Project Organization and Communication (Part 1, Part 2)	Chap 3
Week 5	Project Organization and Communication (ARENA) Requirements Elicitation	Chap 3 & 4
October Week 6	Functional Modeling Analysis (Object Modeling)	Chap 4 & 5
Week 7	Analysis (Dynamic Modeling)	Chap 5
Week 8	***Midterm Exam***	
Week 9	System Design (Decomposing the System)	Chap 6
Week 10	System Design (Addressing Design Goals)	Chap 7
November Week 11	Object Design (Reusing, Design Patterns 1)	Chap 8
Week 12	Object Design (Design Patterns 2, Specifying Interfaces)	Chap 8 & 9

Week 13	Object Design (Object Constraint Language)	Chap 9
Week 14	Mapping Models to Code	Chap 10
December Week 15	Testing, Integration & System Testing	Chap 11
Week 16 & 17	***Final Exam*** (12/18)	

POLICY:

- ***Late Assignment***

  For each assignment, If you submit *m* days later, your maximum grade will 100*(1-0.*m*). For example, if *m*=2, your maximum grade is 100*(1-0.2)=100*0.8=80.

- ***Missed Assignments***

  Discuss with the instructor

- ***Missed Examinations***

  Students who miss an exam should contact the instructor as soon as possible. If it is known in advance that an exam will be missed, the instructor should be contacted before the exam.

- ***Phone Use Policy***

  Phones should be switched off during the mid-term and final exams. Phones, especially smart phones with Internet access and camera, are not allowed to be on person during exams

- ***Electronic Device***

  I have no problem with you using computers or tablets to take notes or consult reference materials during class. Tempting though it may be, please do not check e-mail or visit websites that are not relevant to the course during class. It is a distraction, both for you and (more importantly) for your fellow classmates. Please silence your phones and computers when you enter class.

- ***Important Calendar Dates for university courses***

This course will follow all the university dates published by the university. Students can access those official dates by viewing the university calendar at: https://www.oldwestbury.edu/academics/calendar

- ***Incomplete Grade***

A grade of Incomplete may be assigned when a student has completed and passed a majority of the work required for this course but, for reasons beyond the student's control, cannot complete the entire course. Appropriate documentation should be provided.

Prior to the end of the semester, students must initiate the request for an "I" grade. The incomplete will only be assigned on the basis of an agreement between the instructor and the student (specifying the work to be completed and establishing a general timeline by which the work will be completed.) Incomplete may NOT be resolved by auditing or registering again in a subsequent offering of the course. The date for the completion of the work may be no later than one month before the end of the subsequent semester. Once the work is completed, the instructor will assign an appropriate academic grade.

- ***Academic Integrity***

As members of the Old Westbury community, students are expected to adhere to standards of honesty and ethical behavior. Plagiarism and other types of academic dishonesty are condemned at all academic institutions. These acts detract from the student's intellectual and personal growth by undermining the processes of higher learning and the struggle with one's own expression of ideas and information.

Students are expected to be familiar with the academic standards regarding academic integrity of the University and to uphold the policies in this respect. Students are particularly urged to familiarize themselves with the contents at the website: https://www.oldwestbury.edu/sites/default/files/documents/academic-integrity.pdf and avoid any behavior that could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

- ***Examinations with Disability Services for Students (OSSD)***

If you have or suspect you may have a physical, psychological, medical or learning disability that may impact your course work, please contact Stacey DeFelice, Director, The Office of Services for Students with Disabilities (OSSD), NAB, 2065, Phone: 516-628-5666, Fax (516) 876-3005, TTD: (516) 876-3083. E-mail: defelices@oldwestbury.edu.

The office will help you determine if you qualify for accommodations and assist you with the process of accessing them. All support services are free and all contacts with the OSSD are strictly confidential. SUNY/Old Westbury is committed to assuring that all students have equal access to all learning activities and to social activities on campus.

- ***BASIC NEEDS STATEMENT***

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support at (516)876-3067 or Student Union 303. The College's Panther Food Pantry offers food and personal care items to the campus community in Student Union 301B. The service is anonymous, OW ID is not required. The Pantry website is https://tinyurl.com/yb36bdxc

- ***COUNSELING RESOURCES***

As a college student, there may be times when personal stressors interfere with your academic performance or negatively impact your daily life. If you or someone you know at this college is experiencing mental health challenges, please call the Counseling and Psychological Wellness Services at 516-876-3053 or email us at counselingcenter@oldwestbury.edu .

We're located at the Student Union, Lower Level, Suite 100, and are open Monday to Friday, 9am to 5pm. All services are entirely free and confidential. Students may choose to receive counseling services either in-person, by phone or via video-chat.

If a crisis situation occurs after hours, contact a Residential Director. The University Police, at 516-876-3333, will also be able to offer assistance. The National Suicide Prevention Lifeline also offers help 24/7, and can be contacted at 1-800-273-8255.

- ***TITLE IX, SEXUAL DISCRIMINATION, HARASSMENT AND VIOLENCE***

SUNY Old Westbury prohibits sexual discrimination, harassment and violence, and will promptly respond to all complaints. The purpose of Title IX is to prevent sex discrimination on campus, address reported assaults and incidents, limit the effects of harassment on the educational environment, and prevent its recurrence. If you or someone you know believes they have been subjected to sexual discrimination, harassment or violence, help is available. To report or for more information please visit https://www.oldwestbury.edu/title-ix, please contact the Title IX coordinator, Deputy Title IX coordinator or University Police at 516-876-3333. Confidential resources and support is also available from the counseling professionals in the Counseling & Psychological Wellness Services department, located in the Student Union Lower Level Room LL100 (off the Rotunda) at 516-876-3053.

- **Stephanie Iaccarino, CRC (Faculty/Staff/Student Reports) Deputy Title IX Coordinator Campus Center H-418B Phone: 516-876-2740**

Email: iaccarinos@oldwestbury.edu

- ***Writing Center***

Visit the Writing Center for help brainstorming or organizing your ideas or for feedback on a draft. All services will be offered online for Fall 2020. You can make an appointment for

an online session at https://oldwestbury.mywconline.com. Hours: Mondays and Tuesdays, 11am-8pm and Wednesdays and Thursdays, 10am-7pm. Phone: (516) 876-3093.

- ***Technology and Remote Learning***

Taking online and remote classes can be challenging, and we have several resources to help students adapt.

Blackboard Orientation: Enter Blackboard itself and click on **Student Center** in the upper right corner.
MS Teams: See the MS Teams link on the Old Westbury Tutoring page.
SUNY Online: You can find additional support at https://online.suny.edu/covid19/students/.

For general questions or problems with technology issues, contact the IT Service Desk at servicedesk@oldwestbury.edu or click on the ITS Support Icon on the portal.